



Read the uploaded article on stock price prediction. Accepting the assumptions of this article, design a program that will carryout the concepts of this article, research the best forum sources that today would provide the best market news "on the street" to use to scrape information about the market to process sentiment to inform the prediction machine. Consider forums on Reddit's "Wall Street Bets", X, Yahoo Finance, Discord Channels. Plan a strategy to code a program that will scrape information from the forums, analyze the sentiment of the comments of the stackholders, categorize the stakeholders, weigh the opinions of the most influential stakeholders, and combine this information into market guidance. Explain your rationale for the design of the program, then generate the code for the program so that I can run the program on a local Windows machine running Windows 11. Use Streamlit to generate any UI charts. Take as much time as you need, and if necessary break the problem down into parts, assign parts to separate agents (or subagents) and define your role as the overall orchestrator of the project. When you are done, provide me some instructions on launching the program in a Windows powershell.

I designed a Windows-ready Streamlit program that follows the paper's core assumption: forum participants should be analyzed as distinct stakeholder groups rather than as one blended crowd, and their sentiment should be aggregated with group- and influence-aware weighting before being used as market guidance. I also recommend prioritizing Reddit and X as the best current "on the street" sources, using Discord selectively, and treating Yahoo Finance boards as secondary because they are less robust operationally today. ^[1] ^[2] ^[3] ^[4] ^[5]

Article design

The paper's main contribution is a stakeholder analyzer that combines forum collection, sentiment analysis, and unsupervised stakeholder grouping, then uses group-level measures instead of aggregate forum measures to improve stock-return prediction. Its feature logic is three-part: content features, social-interaction features, and writing-style/activity features; it also uses probabilistic clustering, key phrase extraction, and rolling/sliding windows for near-real-time monitoring. ^[1]

Accepting those assumptions, the modern program should keep the same architecture but update the implementation: collect posts/comments from multiple venues, normalize them into one schema, compute sentiment and subjectivity, infer stakeholder classes, cluster authors into latent groups, weight sentiment by influence, and then export those outputs as factors for a downstream prediction model. That preserves the paper's rationale while adapting it to today's source landscape and developer tooling. ^{[3] [4] [1]}

Source priority

Reddit is the strongest first source because public retail sentiment is abundant there and the developer ecosystem around subreddit collection is mature, with common PRAW/JSON-based workflows widely used in practice. X is the best fast-twitch source for market narrative propagation and influential voices, but the official API is now usage-priced and plan-dependent, so it should be treated as a premium feed with budgeting and query discipline. ^{[2] [6] [3]}

Discord can be highly valuable for niche sector communities or options-centric chatter, but only where your bot is authorized to read channels, and your collector must respect route and global rate limits. Yahoo Finance boards are historically aligned with the paper, but they are weaker as a primary production source today because public access and page structures can be brittle compared with official APIs or stable Reddit collection paths. ^{[4] [7] [5] [1]}

Program strategy

The strategy is to split the system into five modules: collectors, storage, NLP/enrichment, stakeholder modeling, and guidance output. The collectors fetch raw posts from Reddit, X, Yahoo, and Discord into a shared schema; storage writes them into SQLite; enrichment computes sentiment, subjectivity, finance-specific bullish/bearish cues, and stakeholder labels; stakeholder modeling builds author-level features and clusters authors with PCA plus Gaussian mixtures; guidance output produces daily sentiment series, source summaries, stakeholder summaries, and influence-weighted measures for prediction. ^[1]

For stakeholder categorization, use both explicit labels and latent clustering: explicit labels such as "options trader," "fundamental analyst," "technical trader," "macro/regulatory watcher," or "retail momentum" help interpretation, while latent clusters preserve the paper's unsupervised discovery logic for unknown groups. For influence weighting, combine engagement, estimated reach, posting consistency, and thread centrality rather than raw post count alone, because modern platforms differ sharply in how visibility is produced. ^{[5] [3] [1]}

Delivered code

I generated three files for you: a Streamlit app, a requirements file, and a Windows launch/readme guide. The app includes a local SQLite store, Reddit/X/Yahoo connectors, sentiment scoring, stakeholder categorization, author clustering, cluster keyword extraction, influence-weighted sentiment, and Streamlit/Plotly dashboards suitable for local Windows 11 use. ^[2] ^[3] ^[1]

The current code is intentionally a practical starter rather than a finished institutional pipeline: Reddit works through a public JSON path, X uses the official API with your bearer token, Yahoo is treated as opportunistic HTML extraction, and Discord is outlined in the architecture notes for you to implement with server-specific permissions. That design is deliberate because X and Discord now require authenticated, rate-limit-aware collection in ways that should be configured locally with your own credentials and approved access. ^[3] ^[4] ^[5] ^[1]

PowerShell steps

Use these commands on Windows 11 PowerShell to launch the app locally: ^[3]

```
mkdir C:\Users\%env:USERNAME%\Documents\stock_forum_sentiment_app
cd C:\Users\%env:USERNAME%\Documents\stock_forum_sentiment_app
```

Copy the shared files into that folder, then run: ^[3]

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
python -m textblob.download_corpora
streamlit run app.py
```

If PowerShell blocks activation, run this first: ^[3]

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\.venv\Scripts\Activate.ps1
```

The local app should open at <http://localhost:8501> after Streamlit starts. ^[3]

Would you like me to produce the next step as a more complete multi-file production version with dedicated `reddit_connector.py`, `x_connector.py`, `discord_connector.py`, `model.py`, and `config.py` modules?

✱

1. [Zimbra-Stock-Price-Prediction.pdf](#)
2. <https://github.com/RyanElliott10/wsbtickerbot>
3. <https://docs.x.com/changelog>

4. <https://docs.discord.com/developers/reference>
5. <https://support-dev.discord.com/hc/en-us/articles/6223003921559-My-Bot-is-Being-Rate-Limited>
6. <https://docs.x.com/x-api/fundamentals/post-cap>
7. https://developer.yahoo.com/dsp/api/docs/traffic/rate_limits/
8. <https://riomcmahon.me/content/yolo-scraper/>
9. <https://medium.com/swlh/how-to-scrape-large-amounts-of-reddit-data-using-pushshift-1d33bde9286>
10. <https://medium.datadriveninvestor.com/scraping-wallstreetbets-for-stocks-signals-414e63ace210?gi=61f8eadd6691>
11. <https://radiationbox.com/posts/2026-04-30-what-changed-from-the-twitter-api-to-the-x-api.html>